Lösung: Polynom 2. Ordnung

$$f(x) = a_0 + a_1 \cdot x + a_2 x^2$$
$$f'(x) = a_1 + 2a_2 x$$
$$f''(x) = 2a_2$$

$f(0) = -3 \quad \Rightarrow \quad \underline{\underline{a_0 = -3}}$

$f'(0) = 5 \quad \Rightarrow \quad \underline{\underline{a_1 = 5}}$

$f''(x) = 4 \quad \forall x \in \mathbb{R} \quad \Rightarrow \quad 2a_2 = 4 \quad \Rightarrow \quad \underline{\underline{a_2 = 2}}$